Step-by-Step Elastic Kubernetes Service (EKS) Cluster Setup with Auto Mode via AWS Console and eksctl

By

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https://devopstronaut.com/

What is EKS Auto Mode?

EKS Auto Mode is a management enhancement for Amazon Elastic Kubernetes Service (EKS) that simplifies the operational management of Kubernetes clusters and extends AWS's control beyond the cluster itself. It automates the setup and management of the infrastructure required for running workloads, including compute, networking, storage, and scaling, while integrating deeply with AWS services like EC2, EBS, and ELB

Key Features of EKS Auto Mode

1. **Streamlined Management:** Production-ready clusters with minimal operational overhead and automation of key tasks.
2. **Automated Compute and Scaling:**
   - Automated node lifecycle, scaling with Karpenter, and GPU support.
   - Regular node replacements for security (21-day max lifetime).
3. **Networking and Load Balancing:**
   - Automates Pod/service networking with IPv4/IPv6 support.
   - Integrates with AWS Elastic Load Balancers for seamless traffic management.
4. **Cost Optimization:**
   - Automatically consolidates workloads and terminates unused nodes.
   - Supports cost-saving configurations like Spot instances.
5. **Security:**
   - Immutable AMIs, SELinux enforcement, and no direct SSH/SSM access.
   - Regular updates to enhance security posture.
6. **Customizability:**
   - Custom NodePools and NodeClasses for specialized workloads.
   - Support for DaemonSets to add custom monitoring/services.
7. **AWS Integration:** Fully integrates with EC2, EBS, ELB, and other AWS services.

EKS Auto Mode Pricing:

EKS Auto Mode Pricing simplifies Amazon EKS cluster management by automatically selecting and scaling the underlying infrastructure. Pricing includes:

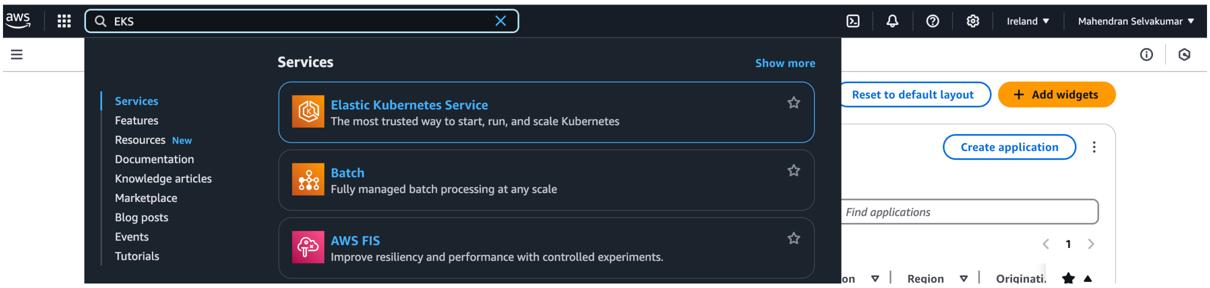
1. EC2 Instance Costs: Standard charges for the EC2 instances provisioned for your workloads.

2. Management Fee: A per-hour fee for each EC2 instance, which varies by instance type, to cover the automation of infrastructure management.

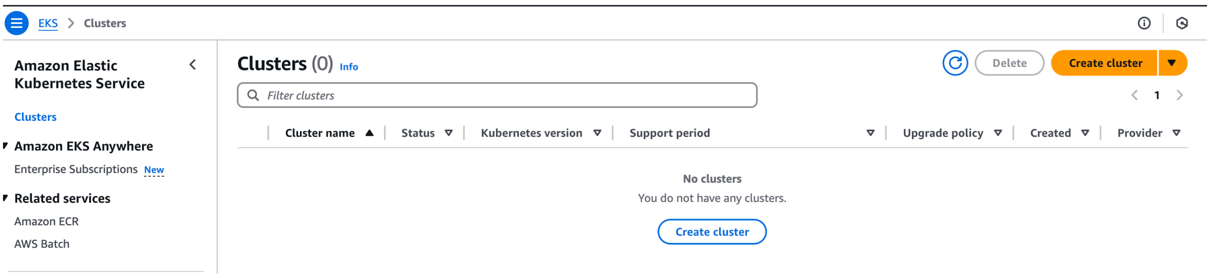
Auto Mode dynamically optimizes resource usage and scales across Availability Zones, ensuring high availability and cost-effectiveness. You only pay for the compute and management fees associated with your active workloads

Step 1: Create EKS Cluster with Auto Mode

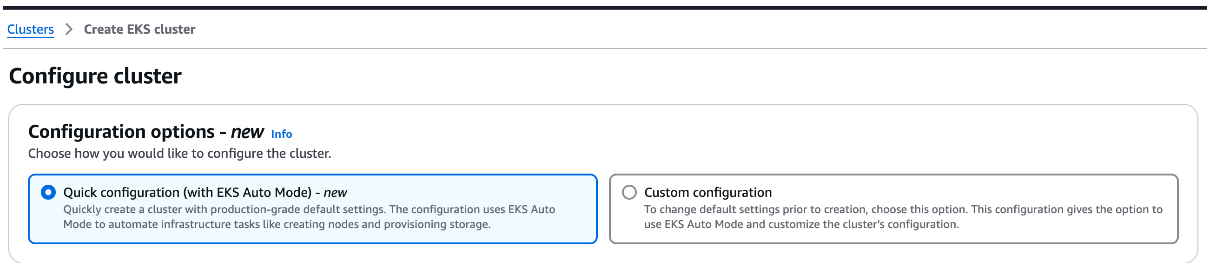
Open the AWS Management Console, search for **EKS**, and click on **Elastic Kubernetes Service**



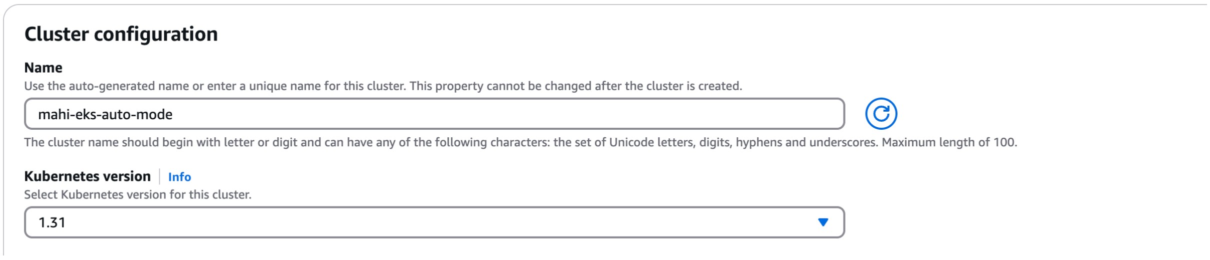
Click on **Create cluster**



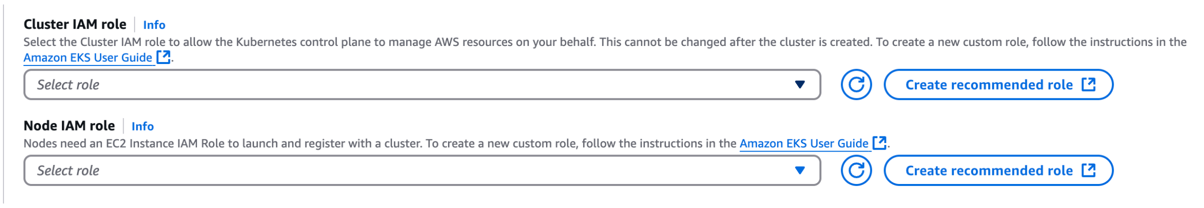
Select **Quick configuration (with EKS Auto Mode)** to enable **EKS Auto Mode**



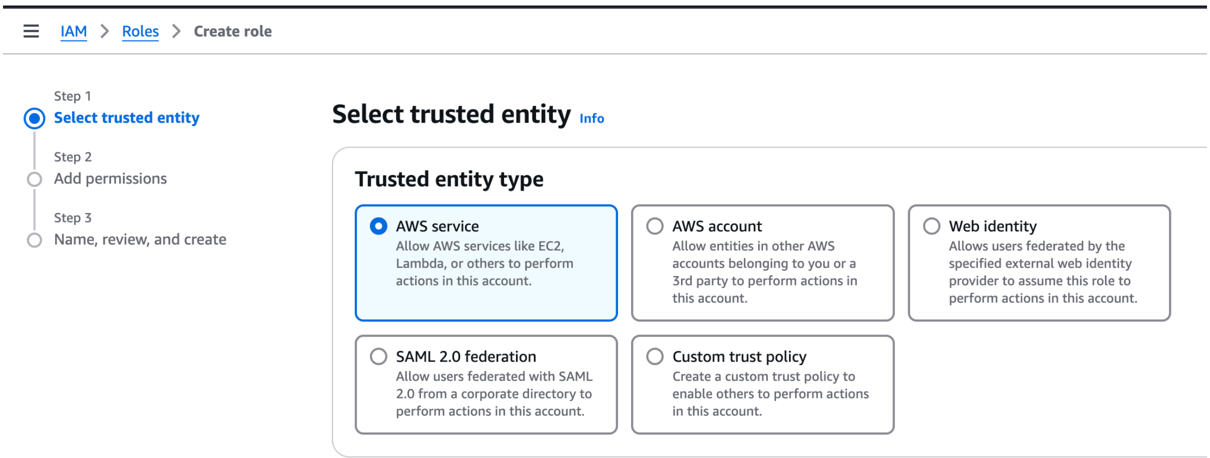
Enter the **EKS cluster name** and select the **Kubernetes version**



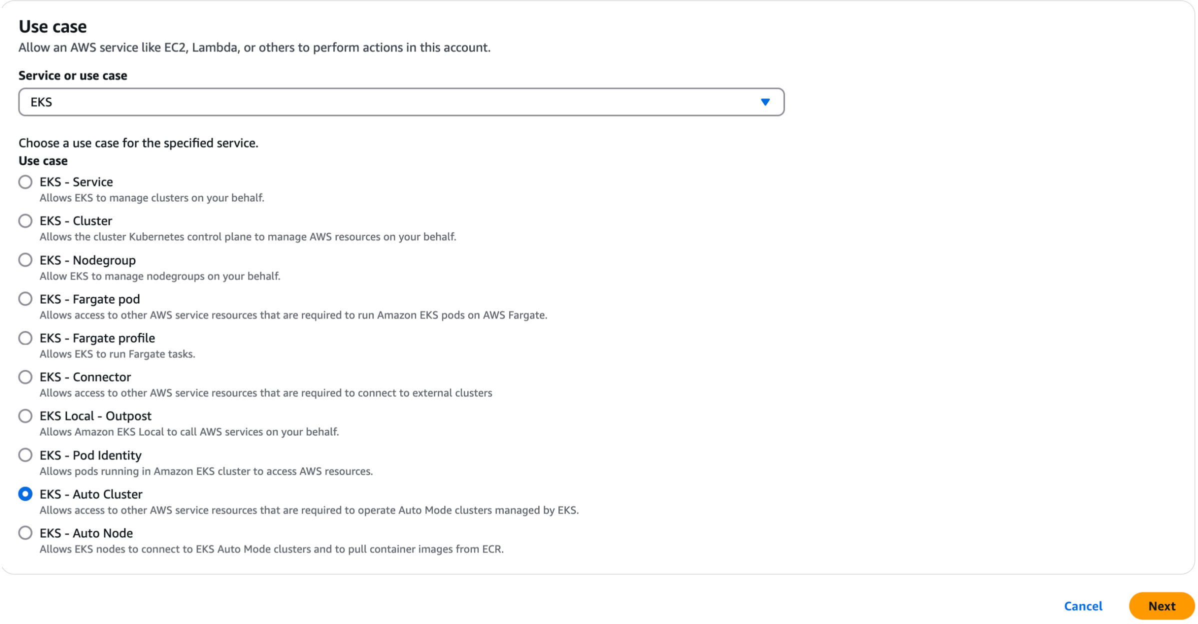
Click **Create recommended role** to create the **Cluster IAM role**



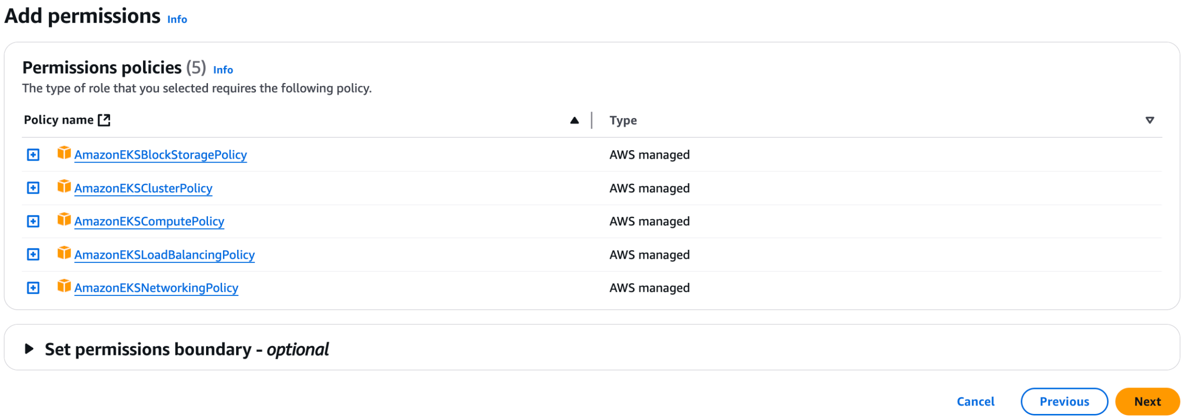
Select **AWS Service** as the trusted entity



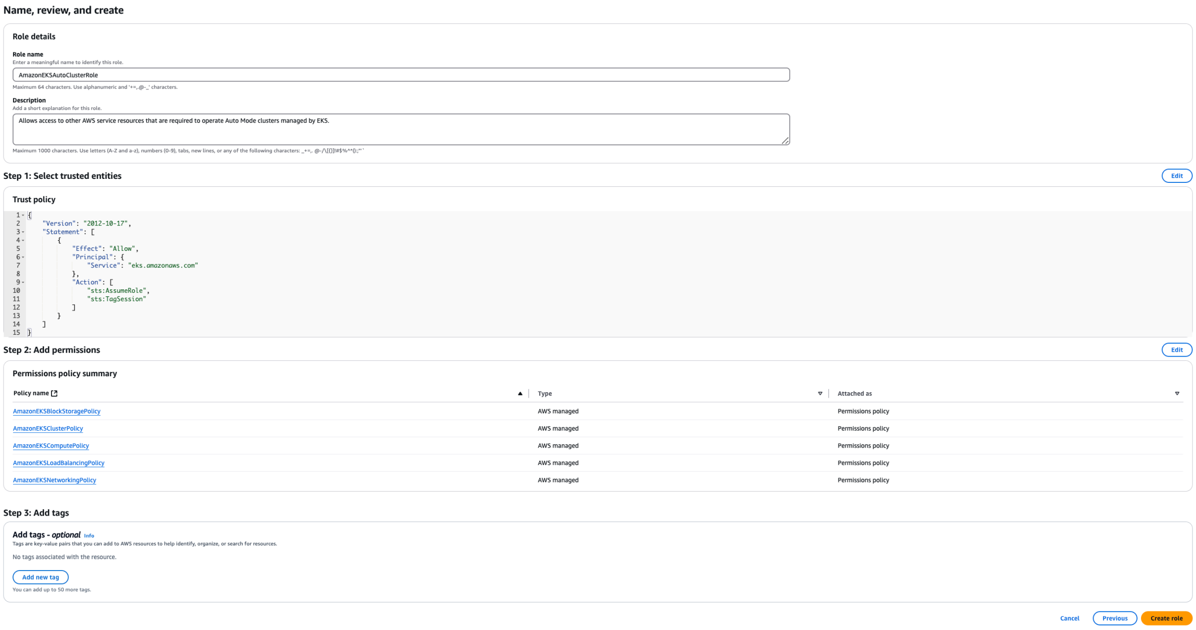
Select **EKS** as the service and **EKS-Auto Cluster** as the use case, then click **Next**



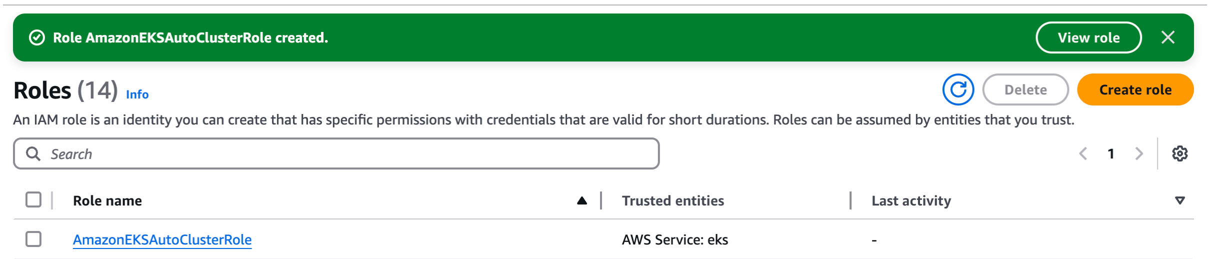
Here, you can view the **permission policies**. After reviewing, click **Next**



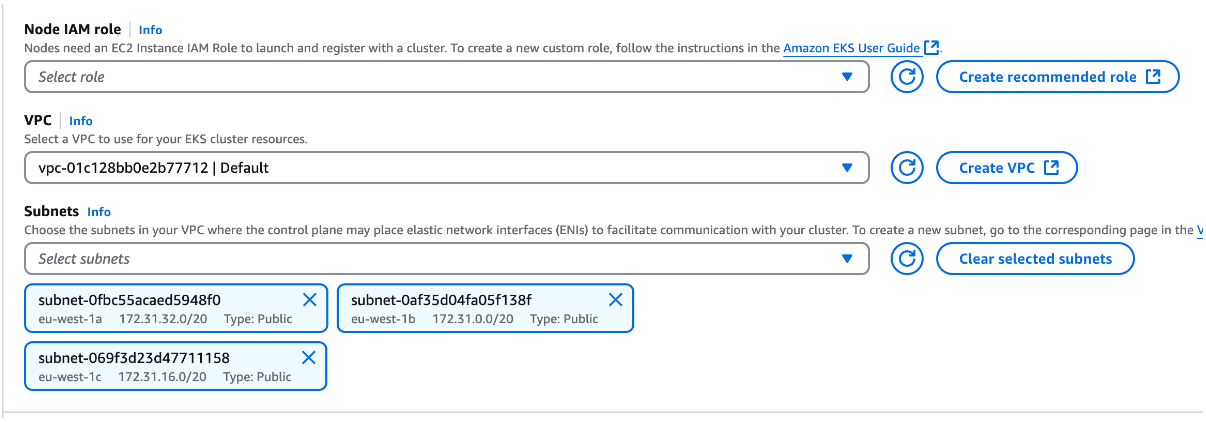
Enter the **role name** and click **Create role**



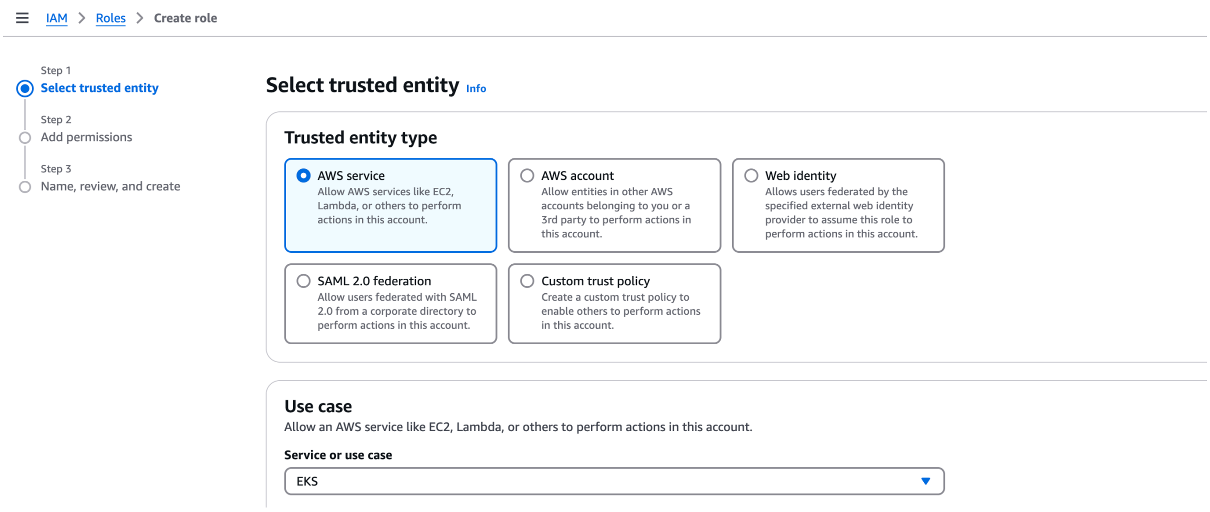
The role has been successfully created for the **EKS Auto Cluster**



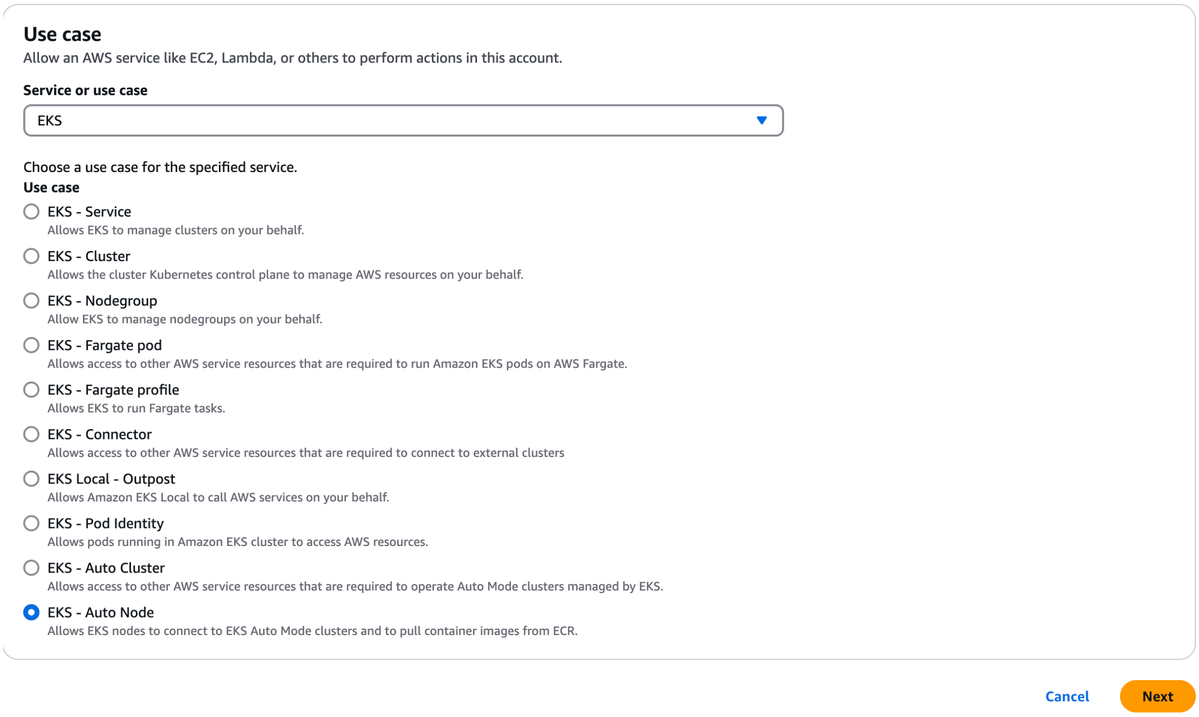
Click **Create recommended role** to create the **Node IAM role**

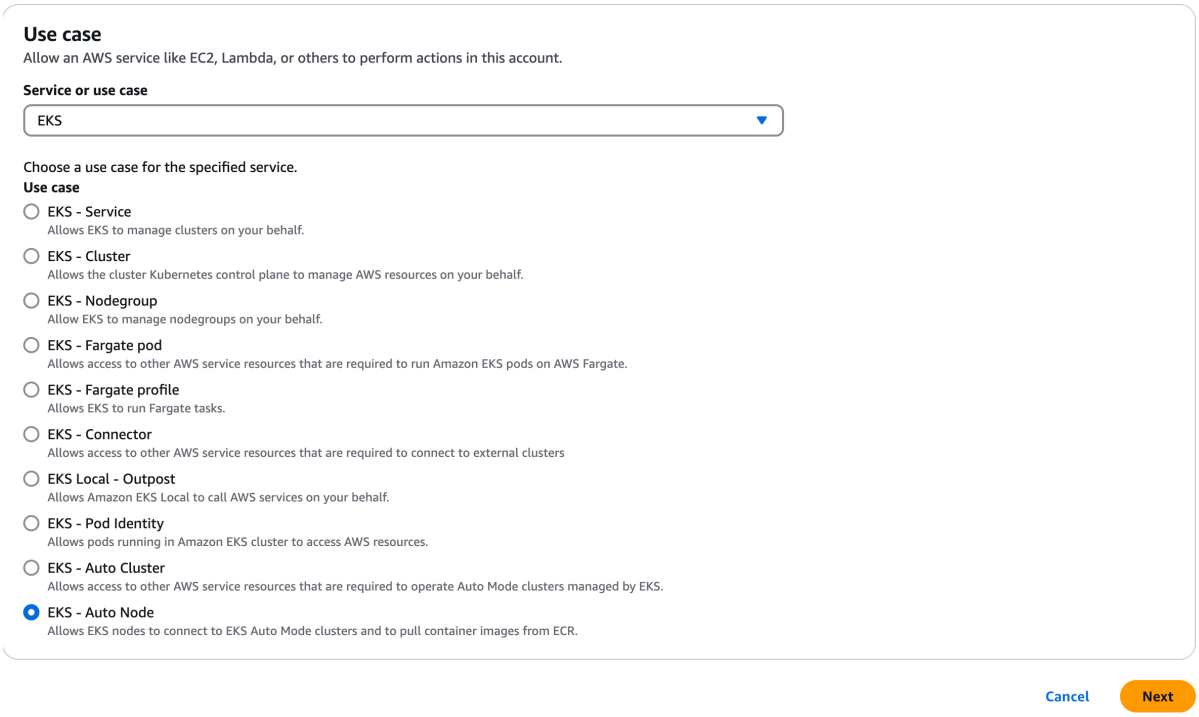


Select **AWS Service** as the trusted entity

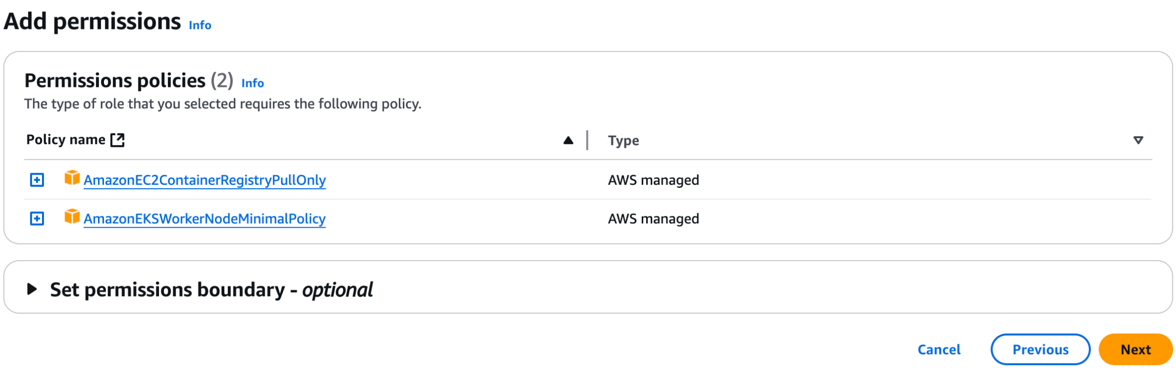


Select **EKS** as the service name and **EKS Auto Node** as the use case

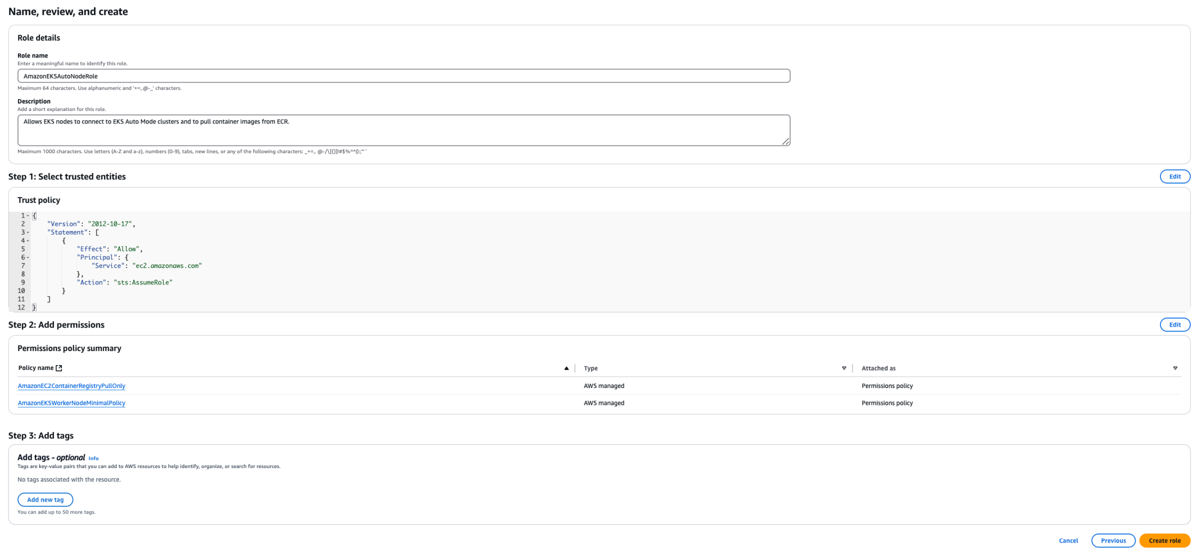




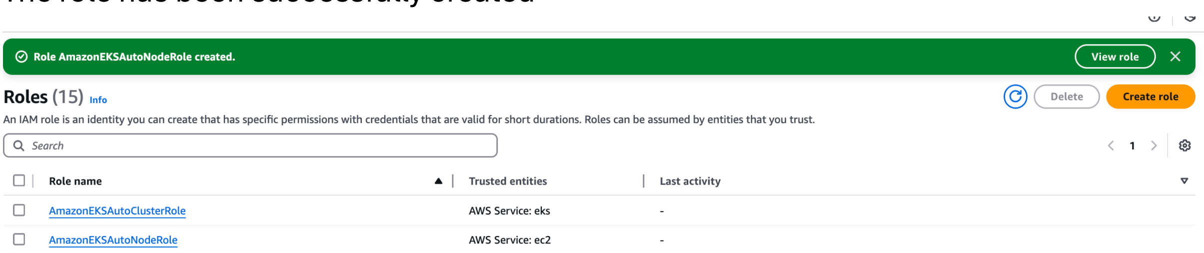
The **permission policies** for **EKS Auto Node** are added. Click **Next** to proceed



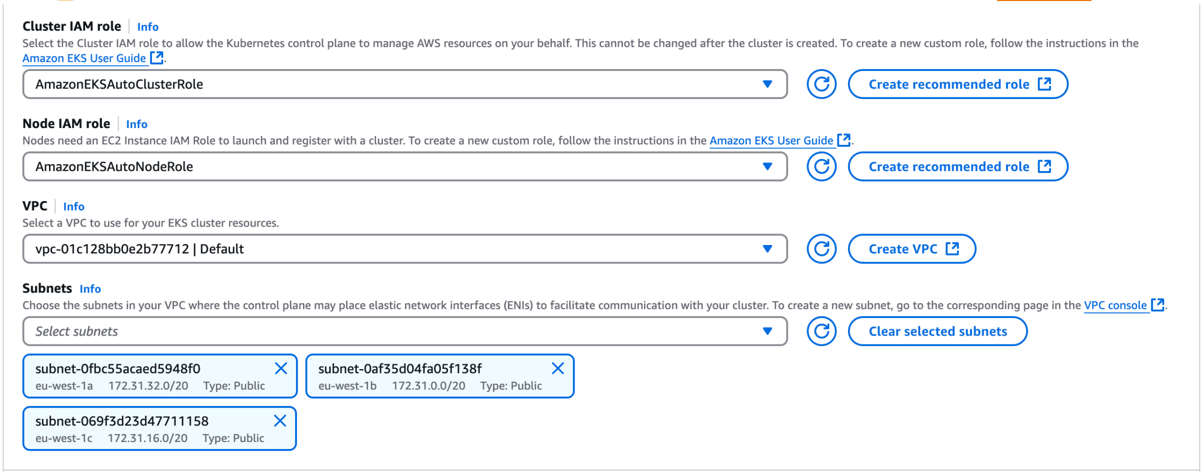
Enter the **role name** and click **Create role**



The role has been successfully created



Refresh the **Node IAM role**, select the role you created, then choose the **VPC** and **Subnets**



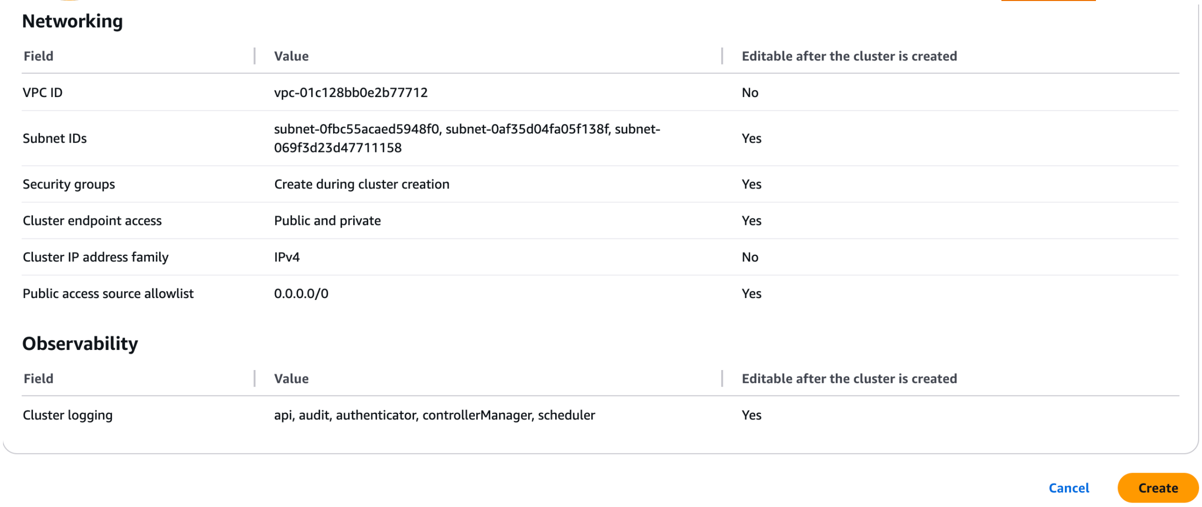
You can now view the **EKS Auto Mode capabilities**

Here, you can see that the **EKS Auto Mode** value is set to **enabled**

Cluster configuration

Field	Value	Editable after the cluster is created
Cluster IAM role	AmazonEKSAutoClusterRole	No
EKS Auto Mode	Enabled View pricing	Yes
Node IAM role	AmazonEKSAutoNodeRole	No
Default node pools	general-purpose, system	Yes
Cluster administrator access	Allow cluster creator cluster administrator access	Yes
Cluster authentication mode	EKS API	No
Secrets encryption	Not enabled	Yes
Upgrade policy	Standard	Yes

Click **Create** to initialize the creation of the cluster


Networking

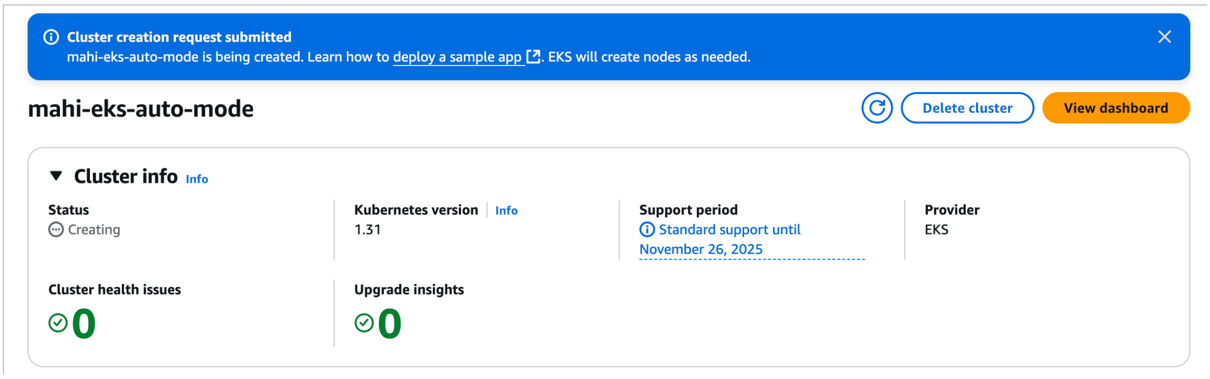
Field	Value	Editable after the cluster is created
VPC ID	vpc-01c128bb0e2b77712	No
Subnet IDs	subnet-0fbc55acaed5948f0, subnet-0af35d04fa05f138f, subnet-069f3d23d47711158	Yes
Security groups	Create during cluster creation	Yes
Cluster endpoint access	Public and private	Yes
Cluster IP address family	IPv4	No
Public access source allowlist	0.0.0.0/0	Yes

Observability

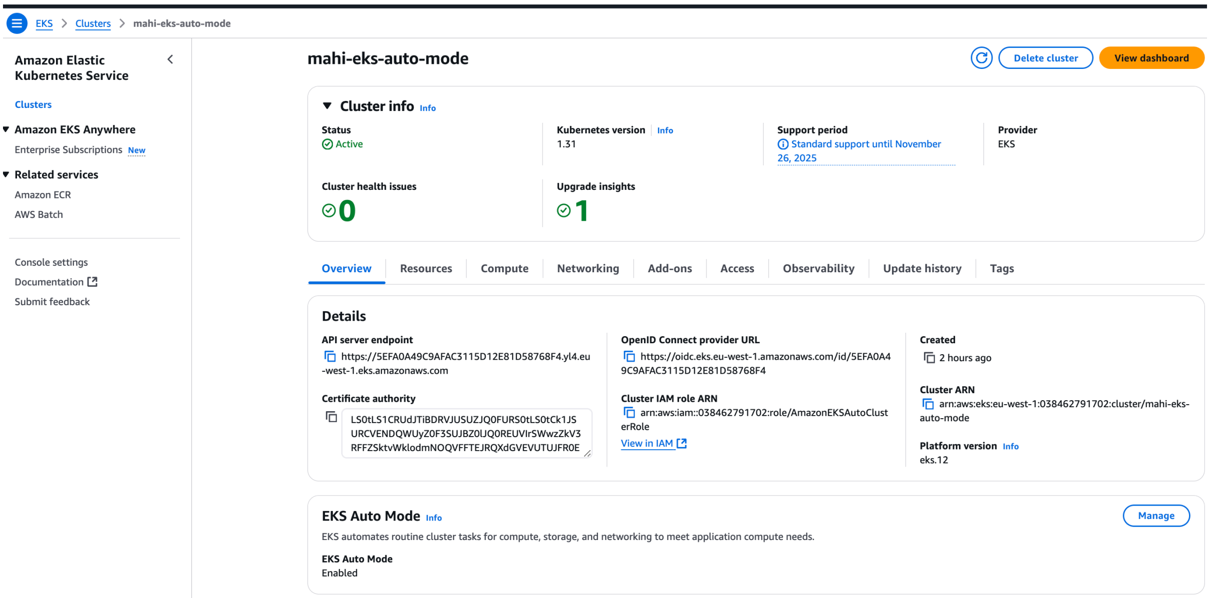
Field	Value	Editable after the cluster is created
Cluster logging	api, audit, authenticator, controllerManager, scheduler	Yes

Cancel Create

The cluster creation request has been submitted, and it will take some time to complete.

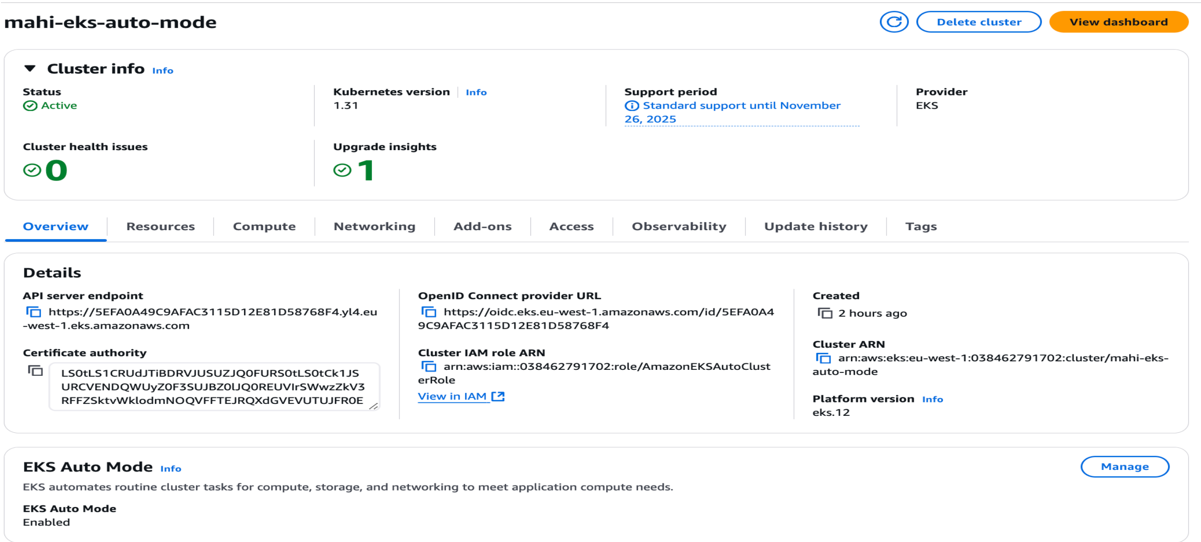


Once the EKS cluster is created, you will see **EKS Auto Mode** displayed as **Enabled**

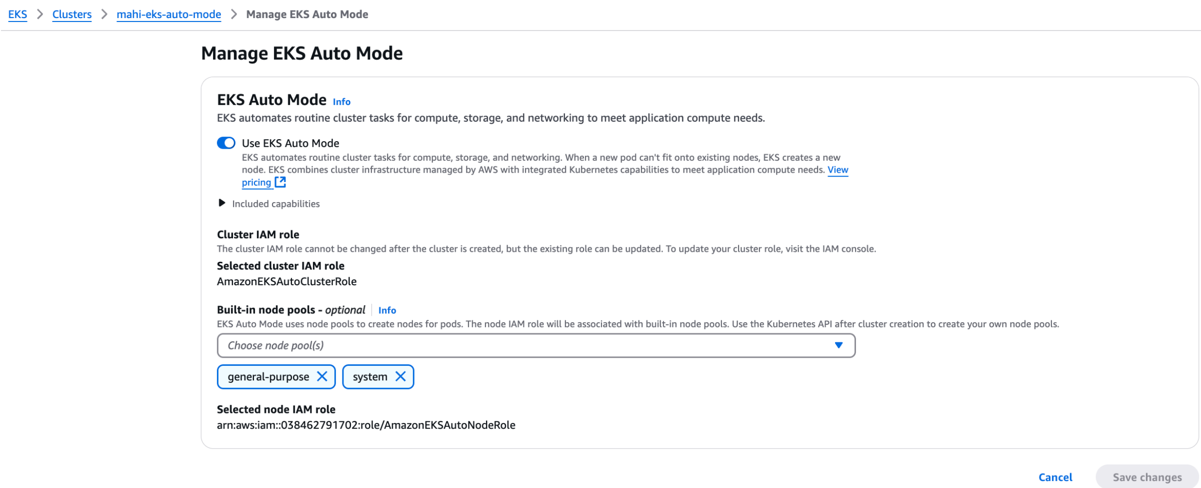


Step 3: Disable EKS Auto Mode

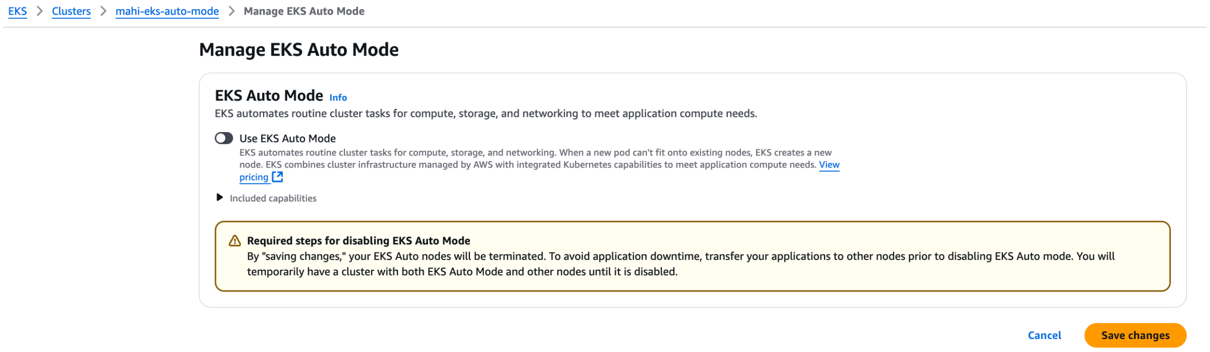
Go to the **EKS Cluster dashboard** and click on **Manager**



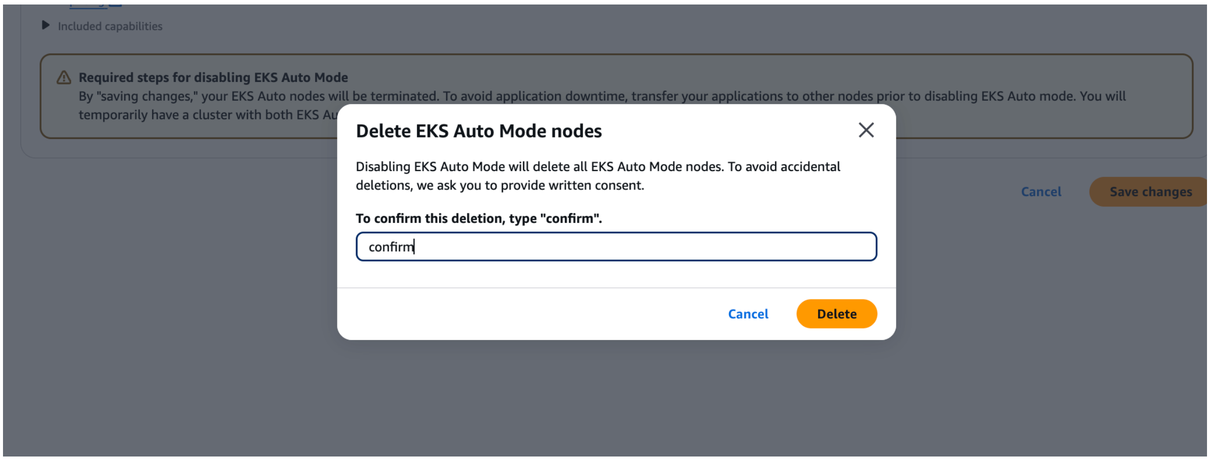
Uncheck the **Use EKS Auto Mode** option to disable



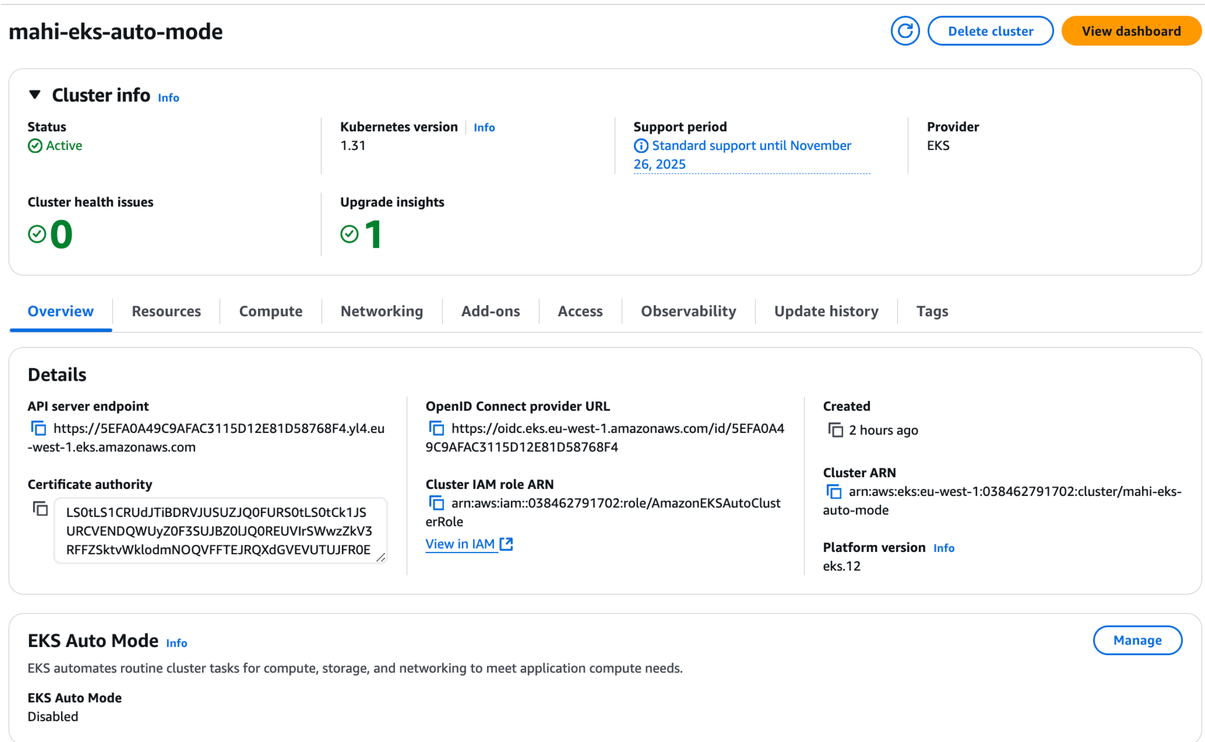
Click **Save changes**



Type **confirm** to delete the EKS Auto Mode nodes, then click **Delete**



Now, you will see the EKS Auto Mode status updated to Disabled



Method 2: Create an EKS Auto Mode Cluster with eksctl

To create an EKS Auto Mode cluster using eksctl, ensure you have the eksctl CLI installed. Then, run this command: **eksctl create cluster --name=devops-auto-mode --enable-auto-mode**. This will create a cluster named devops-auto-mode with EKS Auto Mode enabled

```
mahendranselvakumar@Mahendrans-MBP ~ % eksctl create cluster --name=devops-auto-mode --enable-auto-mode
Error: unknown flag: --enable-auto-mode
```

If you encounter the error **"unknown flag: --enable-auto-mode"**, it means your eksctl version is outdated.

Upgrade the eksctl version to the latest release

Run the **brew update** command to update Homebrew

```
mahendranselvakumar@Mahendrans-MBP ~ % brew update
==> Updating Homebrew...
Updated 4 taps (hashicorp/tap, weaveworks/tap, homebrew/core and homebrew/cask).
==> New Formulae
aliae               azure-core-cpp      eslint_d            libunicode          rshijack
ampl-asl            cargo-chef          eventpp             libxsd-frontend     scooter
ansible@10          cargo-expand        fltk@1.3            lla                 sesh
```

Run the **brew upgrade eksctl** command to upgrade eksctl to the latest version

```
mahendranselvakumar@Mahendrans-MBP ~ % brew upgrade eksctl
==> Upgrading 1 outdated package:
weaveworks/tap/eksctl 0.190.0 -> 0.196.0
==> Fetching dependencies for weaveworks/tap/eksctl: aws-iam-authenticator and kubernetes-cli
==> Fetching aws-iam-authenticator
==> Downloading https://ghcr.io/v2/homebrew/core/aws-iam-authenticator/manifests/0.6.28
######################################################################################################################## 100.0%
==> Downloading https://ghcr.io/v2/homebrew/core/aws-iam-authenticator/blobs/sha256:2247afcae2b6628195756935701bc46d873d98ae8
######################################################################################################################## 100.0%
==> Fetching kubernetes-cli
==> Downloading https://ghcr.io/v2/homebrew/core/kubernetes-cli/manifests/1.31.3
######################################################################################################################## 100.0%
==> Downloading https://ghcr.io/v2/homebrew/core/kubernetes-cli/blobs/sha256:118193fcf9a4f91c21f0583589388ae6dc287963e591642f
######################################################################################################################## 100.0%
==> Fetching weaveworks/tap/eksctl
==> Downloading https://github.com/eksctl-io/eksctl/releases/download/v0.196.0/eksctl_Darwin_arm64.tar.gz
==> Downloading from https://objects.githubusercontent.com/github-production-release-asset-2e65be/134539560/9e30a5bb-0b14-45c
######################################################################################################################## 100.0%
==> Upgrading weaveworks/tap/eksctl
  0.190.0 -> 0.196.0
==> Installing dependencies for weaveworks/tap/eksctl: aws-iam-authenticator and kubernetes-cli
==> Installing weaveworks/tap/eksctl dependency: aws-iam-authenticator
==> Downloading https://ghcr.io/v2/homebrew/core/aws-iam-authenticator/manifests/0.6.28
Already downloaded: /Users/mahendranselvakumar/Library/Caches/Homebrew/downloads/d17a9e218d5379106bf572bd9994bd2cca2ed29327277aeddab5af351236d643--aws-iam-authenticat
or-0.6.28.bottle_manifest.json
==> Pouring aws-iam-authenticator--0.6.28.arm64_sonoma.bottle.tar.gz
🍺  /opt/homebrew/Cellar/aws-iam-authenticator/0.6.28: 7 files, 59.1MB
==> Installing weaveworks/tap/eksctl dependency: kubernetes-cli
==> Downloading https://ghcr.io/v2/homebrew/core/kubernetes-cli/manifests/1.31.3
Already downloaded: /Users/mahendranselvakumar/Library/Caches/Homebrew/downloads/f8fd19d10e239038f339af3c9b47978cb154932f089fbf6b7d67ea223df378de--kubernetes-cli-1.31
.3.bottle_manifest.json
==> Pouring kubernetes-cli--1.31.3.arm64_sonoma.bottle.tar.gz
🍺  /opt/homebrew/Cellar/kubernetes-cli/1.31.3: 237 files, 60.2MB
==> Installing weaveworks/tap/eksctl
==> Caveats
zsh completions have been installed to:
  /opt/homebrew/share/zsh/site-functions
==> Summary
🍺  /opt/homebrew/Cellar/eksctl/0.196.0: 6 files, 155.3MB, built in 3 seconds
==> Running `brew cleanup eksctl`...
Disable this behaviour by setting HOMEBREW_NO_INSTALL_CLEANUP.
Hide these hints with HOMEBREW_NO_ENV_HINTS (see `man brew`).
Removing: /opt/homebrew/Cellar/eksctl/0.190.0... (6 files, 151MB)
==> Caveats
```

Now, run the command **eksctl create cluster --name=devops-auto-mode --enable-auto-mode** to create the EKS Auto Mode cluster

```
mahendranselvakumar@Mahendrans-MBP ~ % eksctl create cluster --name=devops-auto-mode --enable-auto-mode
2024-12-02 23:02:06 [ℹ]  eksctl version 0.196.0
2024-12-02 23:02:06 [ℹ]  using region eu-west-1
2024-12-02 23:02:06 [ℹ]  setting availability zones to [eu-west-1a eu-west-1c eu-west-1b]
2024-12-02 23:02:06 [ℹ]  subnets for eu-west-1a - public:192.168.0.0/19 private:192.168.96.0/19
2024-12-02 23:02:06 [ℹ]  subnets for eu-west-1c - public:192.168.32.0/19 private:192.168.128.0/19
2024-12-02 23:02:06 [ℹ]  subnets for eu-west-1b - public:192.168.64.0/19 private:192.168.160.0/19
2024-12-02 23:02:06 [ℹ]  using Kubernetes version 1.30
2024-12-02 23:02:06 [ℹ]  creating EKS cluster "devops-auto-mode" in "eu-west-1" region with
2024-12-02 23:02:06 [ℹ]  if you encounter any issues, check CloudFormation console or try 'eksctl utils describe-stacks --region=eu-west-1 --cluster=devops-auto-mode'
2024-12-02 23:02:06 [ℹ]  Kubernetes API endpoint access will use default of {publicAccess=true, privateAccess=false} for cluster "devops-auto-mode" in "eu-west-1"
2024-12-02 23:02:06 [ℹ]  CloudWatch logging will not be enabled for cluster "devops-auto-mode" in "eu-west-1"
2024-12-02 23:02:06 [ℹ]  you can enable it with 'eksctl utils update-cluster-logging --enable-types={SPECIFY-YOUR-LOG-TYPES-HERE (e.g. all)} --region=eu-west-1 --cluster=devops-auto-mode'
2024-12-02 23:02:06 [ℹ]
2 sequential tasks: { create cluster control plane "devops-auto-mode", wait for control plane to become ready
}
2024-12-02 23:02:06 [ℹ]  building cluster stack "eksctl-devops-auto-mode-cluster"
2024-12-02 23:02:06 [ℹ]  deploying stack "eksctl-devops-auto-mode-cluster"
2024-12-02 23:02:36 [ℹ]  waiting for CloudFormation stack "eksctl-devops-auto-mode-cluster"
2024-12-02 23:03:06 [ℹ]  waiting for CloudFormation stack "eksctl-devops-auto-mode-cluster"
2024-12-02 23:04:07 [ℹ]  waiting for CloudFormation stack "eksctl-devops-auto-mode-cluster"
2024-12-02 23:05:07 [ℹ]  waiting for CloudFormation stack "eksctl-devops-auto-mode-cluster"
2024-12-02 23:06:07 [ℹ]  waiting for CloudFormation stack "eksctl-devops-auto-mode-cluster"
2024-12-02 23:07:07 [ℹ]  waiting for CloudFormation stack "eksctl-devops-auto-mode-cluster"
2024-12-02 23:08:07 [ℹ]  waiting for CloudFormation stack "eksctl-devops-auto-mode-cluster"
2024-12-02 23:09:07 [ℹ]  waiting for CloudFormation stack "eksctl-devops-auto-mode-cluster"
2024-12-02 23:10:08 [ℹ]  waiting for CloudFormation stack "eksctl-devops-auto-mode-cluster"
2024-12-02 23:11:08 [ℹ]  waiting for CloudFormation stack "eksctl-devops-auto-mode-cluster"
2024-12-02 23:12:08 [ℹ]  waiting for CloudFormation stack "eksctl-devops-auto-mode-cluster"
2024-12-02 23:13:08 [ℹ]  waiting for CloudFormation stack "eksctl-devops-auto-mode-cluster"
2024-12-02 23:15:09 [ℹ]  waiting for the control plane to become ready
2024-12-02 23:15:10 [✔]  saved kubeconfig as "/Users/mahendranselvakumar/.kube/config"
2024-12-02 23:15:10 [ℹ]  no tasks
2024-12-02 23:15:10 [✔]  all EKS cluster resources for "devops-auto-mode" have been created
2024-12-02 23:15:11 [ℹ]  kubectl command should work with "/Users/mahendranselvakumar/.kube/config", try 'kubectl get nodes'
2024-12-02 23:15:11 [✔]  EKS cluster "devops-auto-mode" in "eu-west-1" region is ready
mahendranselvakumar@Mahendrans-MBP ~ %
```

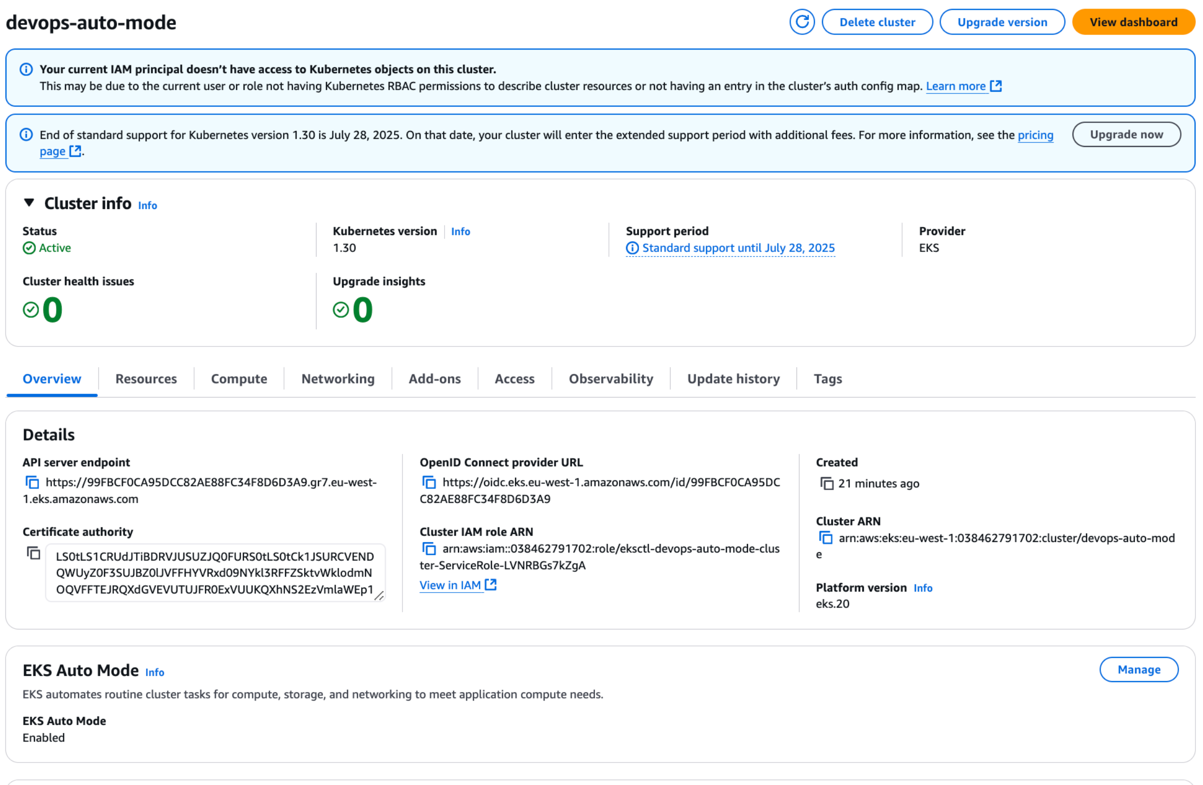
You will be able to view the newly created EKS cluster in the AWS Management Console

Clusters (2) Info

Delete | Create cluster

	Cluster name	Status	Kubernetes version	Support period	Upgrade policy	Created	Provider
◉	devops-auto-mode	Active	1.30 Upgrade now	Standard support until July 28, 2025	Extended	20 minutes ago	EKS
○	mahi-eks-auto-mode	Active	1.31	Standard support until November 26, 2025	Standard	3 hours ago	EKS

Navigate to the newly created EKS cluster in the AWS Management Console, where you will see that **EKS Auto Mode** is set to **Enabled**



Disabling **EKS Auto Mode** is not possible using the eksctl command. To remove Auto Mode, you need to delete the existing cluster and recreate it without enabling Auto Mode. To delete the cluster, run this command **eksctl delete cluster --name=devops-auto-mode**

```
mahendranselvakumar@Mahendrans-MBP ~ % eksctl delete cluster --name=devops-auto-mode
2024-12-02 23:27:54 [ℹ]  deleting EKS cluster "devops-auto-mode"
2024-12-02 23:27:54 [ℹ]  deleted 0 Fargate profile(s)
2024-12-02 23:27:54 [✔]  kubeconfig has been updated
2024-12-02 23:27:54 [ℹ]  cleaning up AWS load balancers created by Kubernetes objects of Kind Service or Ingress
2024-12-02 23:27:55 [ℹ]  1 task: { delete cluster control plane "devops-auto-mode" [async] }
2024-12-02 23:27:55 [ℹ]  will delete stack "eksctl-devops-auto-mode-cluster"
2024-12-02 23:27:56 [✔]  all cluster resources were deleted
mahendranselvakumar@Mahendrans-MBP ~ %
```

Conclusion

EKS Auto Mode simplifies Kubernetes cluster management by automating infrastructure and operational tasks, making it easier to deploy and maintain production-ready environments. With features like automated scaling, cost optimization, enhanced security, and seamless AWS integration, it provides a balanced approach to managing workloads effectively while reducing operational overhead.

Keep Learning, Keep EKS-ing!!

Feel free to reach out to me, if you have any other queries or suggestions Stay connected on LinkedIn: Mahendran Selvakumar

Stay connected on Medium: https://devopstronaut.com/